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Real-Time Drilling Performance Monitoring Using AI-Enhanced Microservices and LLM Integration

H. Sahebi and D. Sui, Department of Energy and Petroleum Engineering, University of Stavanger, Norway

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Abstract

This paper presents a modular, AI-driven platform for real-time drilling performance monitoring, developed to enhance operational decision-making and reduce non-productive time. Built on a microservice architecture, the system integrates both real-time and simulated data sources, including a drilling simulator and a mock application programming interface (API) that replays historical field data as if it were live. This hybrid approach enables controlled, repeatable testing while supporting model development, validation, and demonstration in realistic environments. Continuous learning is achieved through multi-model digital twins that update in real time, fine-tuning pre-trained models without restarting from scratch. Multiple supervised and temporal machine learning models run in parallel, each dedicated to monitoring specific drilling parameters such as weight on bit, rate of penetration, torque, and hookload. A visualization dashboard provides real-time comparison of predicted and measured values, along with anomaly detection based on deviation thresholds and trend shifts. Large language model (LLM) agents are incorporated to process drilling reports, extract structured parameters, and support natural language queries through a chat-based interface. Demonstrations using simulator data and replayed field datasets confirm the system's capability to integrate diverse AI components for monitoring and scenario testing. These results indicate that the proposed architecture provides a flexible and scalable foundation for deploying AI-enhanced drilling performance systems in operational environments, with potential to accelerate adoption of advanced analytics in the drilling industry.

Keywords: Microservice architecture, Continuous training while drilling, Stuck pipe monitoring, Large language model agent for command/report generation, Vector database

Introduction

Modern drilling operations generate large volumes of real-time data, requiring intelligent systems capable of analyzing, interpreting, and reacting to changing conditions with minimal delay. Traditional monolithic software often falls short in handling the scale and complexity of such data. In response, this paper presents an AI-enhanced, microservice-based platform for real-time drilling performance monitoring, built to support continuous data processing, machine learning, and advanced user interaction through large language models.

The system architecture is composed of modular, containerized microservices that isolate key functions such as data ingestion, simulation, prediction, and user-facing services. At the current development stage, the platform integrates two primary sources of real-time data—historic field datasets replayed via a mock API and simulated data from a drilling simulator—to demonstrate the system's capabilities. These sources are used to train and validate base models offline, enabling realistic performance testing without relying on direct access to operational drilling data, which is often difficult to obtain. In live operations, the system would select an appropriate pre-trained base model for each key parameter—such as surface torque, weight on bit, and standpipe pressure—and run these models separately but in parallel. This multi-model setup functions as a digital twin, predicting expected parameter behavior, comparing it with actual sensor readings, and triggering continuous fine-tuning to adapt to changing drilling conditions without training from scratch.

AI models embedded within dedicated microservices are continuously trained using incoming data, enabling real-time predictions of key drilling parameters and anomaly detection. A digital twin approach has been applied for early stuck pipe detection, with preliminary results indicating promising performance, though formal validation is ongoing. In addition, the platform incorporates generative AI capabilities through LLM agents and prompt engineering techniques. These agents are designed to assist with tasks such as report generation, contextual chat support for operators, and intelligent text-based interaction with the system. A key application of this is in LLM-assisted drill string dynamic simulation, where agents extract drill string specifications and control parameters directly from unstructured drilling reports and automatically configure simulation inputs, and enhancing accessibility. This paper presents the system architecture, integration strategy, and early performance results. It highlights how the combination of microservices, continuous AI training, and LLM agents—enabled through prompt engineering—can deliver scalable, intelligent, and user-friendly tools for real-time drilling operations.

System Architecture and Data Integration

In our earlier work, Sahebi and Sui (Sahebi and Sui 2024) introduced a modular, microservice-based software platform for drilling performance monitoring and prediction using AI components deployed in containerized environments. That study demonstrated the benefits of microservices in achieving flexible integration of real-time data pipelines and continuous machine learning in drilling applications. The architecture developed in that earlier work forms the foundation of the present study, which extends it to include large language models (LLMs) and simulation-driven modules.

AI Models and Digital Twin for Stuck Pipe Detection

In parallel, Sui and Sahebi (Sui and Sahebi 2023) proposed an automatic method for trend and dynamic analysis of temporal drilling data using Qualitative Trend Analysis (QTA). This earlier study highlighted the value of trend extraction for drilling diagnostics and introduced a model evaluation approach that incorporates trend alignment in addition to conventional error metrics. This previously developed method underpins the trend-tracking logic in the present system and motivates future development of AI components with more interpretable behavior.

Machine learning has become a key enabler in drilling performance prediction, with applications in rate of penetration (ROP) forecasting, stuck pipe risk detection, and drilling state classification. Early work by Esmaeli et al. (Esmaeili et al. 2012) and later by Hegde et al. (Hegde et al. 2018) demonstrated the effectiveness of ML models—such as support vector machines, decision trees, and neural networks—in predicting ROP based on surface and downhole parameters. More recent efforts have incorporated deep learning approaches like Long Short-Term Memory (LSTM) to better capture temporal dynamics in drilling data streams. However, most existing models are trained offline and applied in fixed settings, limiting their adaptability across formations and operations. Challenges related to generalizability, model transparency, and real-time integration remain active research areas.

While stuck pipe events are among the most costly and disruptive incidents in drilling operations, in this study they are used only as an example to demonstrate the capabilities of the proposed multi-model digital twin approach. Traditional detection methods, based on static thresholds and rule-based logic, are limited in handling complex and time-varying downhole conditions. Recent work has explored data-driven approaches—such as machine learning classifiers and time-series anomaly detection—to identify early signs of stuck pipe risks using surface measurements like torque, hookload, and weight on bit (Al-Dushaishi et al. 2021). However, these models often require labeled incident data and have limited transferability across different wells or formations. The same multi-model digital twin framework presented here can be applied to other types of operational anomalies, making it broadly relevant beyond stuck pipe detection.

To address these challenges, digital twin methodologies have been introduced to simulate ideal mechanical behavior in real time. By comparing simulated and actual behavior, deviations can signal emerging dysfunctions (Liu et al. 2024). In our system, we adopt this concept through a predictive digital twin model that estimates expected mechanical responses based on live drilling parameters. Although full validation is ongoing, early results demonstrate promising alignment between anomalies and deviations, forming a basis for real-time advisory and early warning systems.

LLMs and Generative AI in Drilling Applications

LLMs and generative AI have introduced new possibilities for improving automation, interpretability, and human–machine interaction in engineering systems. In technical domains, they are increasingly used for intelligent document processing, contextual search, and interactive reporting (Brown et al. 2020; Bommasani et al. 2021). Although their application in drilling operations is still emerging, their ability to process unstructured text and deliver natural language responses offers clear potential for operator support and data-driven decision-making.

In this study, we incorporate LLM-based agents—implemented using the OpenAI API (Paredes et al. 2023) and Lang Chain framework (Jeong 2023)—as an integral part of the platform's automation and user interaction layer. These agents extract drill string parameters and control inputs from unstructured daily drilling reports, converting them into structured formats for direct use in the simulation modules. Retrieval-augmented generation (RAG) (Lewis et al. 2020) with a vector database enables context-aware semantic search over drilling documents, allowing the system to ground responses in domain-specific knowledge. Through a chat-based interface, engineers can query data, request parameter summaries, or auto-generate report content in real time. This functionality directly supports the platform's goal of reducing manual data handling and enabling natural language access to complex engineering workflows, marking an early but practical step toward operationalizing Generative AI (GenAI) (Bommasani et al. 2021) in real-time drilling environments.

Methodology

Microservice-Based System Architecture

The platform is built on a modular microservice architecture designed for real-time drilling performance monitoring. Each key function—data ingestion, simulation, AI inference, and LLM interaction—is encapsulated in its own container, enabling isolated development, deployment, and scaling. Services communicate via Representational State Transfer Application Programming Interfaces (REST APIs) (Masse 2011) or message queues, ensuring asynchronous and resilient data exchange.

Fig.1. illustrates the modular architecture of the proposed real-time drilling performance monitoring platform. The system is composed of containerized microservices that manage different functional layers, including data ingestion, user administration, real-time model training, feature engineering, and generative AI interaction. Real-time drilling data is streamed through an API (1) and stored in a dedicated database (C1), while user access and system control are managed through a lightweight API (2) and user database

(C2). Machine learning workflows are coordinated by a central API (3), which interacts with separate servers for real-time training (3.1) and feature calculation (3.2). Trained models are distributed across parallel workers via a Redis-based (Remote Dictionary Server) (Sanfilippo 2025) task queue, allowing for scalable and distributed inference and model updating. A key innovation is the integration of a generative AI module (4), which supports intelligent parameter extraction for the drill string dynamic setup, and user interaction using LLM agents. This design enables a flexible and extensible AI platform that can adapt to a variety of drilling scenarios in real time.

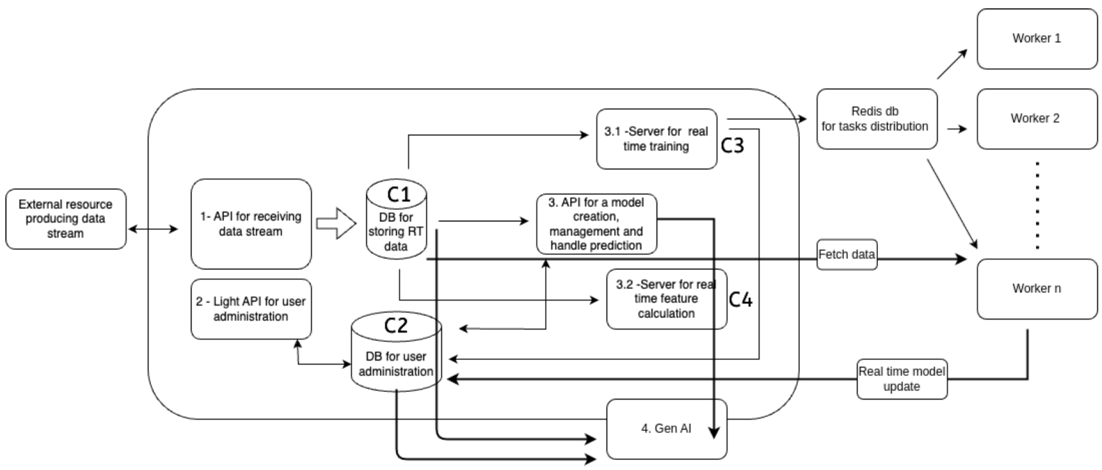

Figure 1—Modular microservice architecture of the real-time drilling performance monitoring platform. Key components include data ingestion APIs, real-time training and feature calculation servers, a Redis-based (Sanfilippo 2025) task distribution mechanism for parallel ML workers, and a Generative AI module for intelligent user interaction and automation. All services are containerized and interact through RESTful APIs.

Core components:

- **User Service**: Interfaces with operators and engineers via API endpoints.
- **Data Receiver**: Streams and preprocesses data from simulators or mock APIs.
- **AI Model Service**: Performs machine learning inference and supports continuous training.
- **LLM Agent Service**: Handles prompt-based tasks such as data extraction and automated report generation.

To support parallel machine learning development, the platform includes a distributed training and inference setup using multiple worker processes coordinated through a Redis server (Sanfilippo 2025) in-memory data store. This architecture enables scalable task distribution, real-time queuing, and efficient model updates without service interruption. The containerized structure also allows for secure isolation of sensitive components, such as digital twin models or LLM logic, enhancing both data integrity and operational flexibility. This design forms the backbone for integrating Generative AI capabilities and real-time prediction services, discussed in the next section.

Data Integration and Simulation

To enable real-time and reproducible testing of AI models, the platform integrates two primary data sources: a live connection to a web-based drilling simulator (Saadallah et al. 2019) and a mock API that replays historic Volve field data (Tunkiel et al. 2020) as a real-time stream. The simulator provides a browser-accessible environment tailored for research, education, and innovation in drilling technology. In this work, it enables bidirectional communication between the microservice architecture and the simulator—facilitating real-time data streaming and control signal transmission. The Volve dataset, released by a major

Norwegian operator, provides detailed drilling, production, and reservoir data from the Volve field in the North Sea, active between 2008 and 2016 (Equinor 2018). It serves as a benchmark for research in data-driven drilling and reservoir modeling. Both sources feed data into the system through the Data Receiver microservice, where they are cleaned, aligned, and normalized for further processing.

The web-based drilling simulator (Saadallah et al. 2019) integration uses RESTful APIs (Masse 2011) to stream high-frequency drilling parameters such as weight on bit, torque, RPM, and depth. This allows the platform to test its real-time inference and training capabilities under controlled but realistic conditions. The mock Volve API is designed to simulate actual drilling operations using historic time-series data. By replaying this data at configurable speeds, the system can emulate real-time behavior while enabling regression testing, pipeline tuning, and benchmarking of AI models. Data from both sources is timestamped and preprocessed to a unified format before being passed to the AI Model Service and stored for training, prediction, or user interface display. This dual-source design ensures that the platform is suitable for both live deployment scenarios and offline development environments, supporting iterative improvement of AI models under real-world constraints.

AI Model Training and Inference Pipeline

Model Architecture and Features. The system currently supports supervised learning models and temporal models like Long Short-Term Memory (LSTM). Input features include surface and downhole parameters such as weight on bit, rate of penetration, torque, RPM, and hook load. These features are derived from both a web-based drilling simulator (Saadallah et al. 2019) and replayed Volve field data (Equinor 2018).

Continuous Training. To maintain model relevance, a sliding window approach is used for continuous training. Data is collected in batches, preprocessed, and used to incrementally retrain models at regular intervals. This ensures adaptability to changing formation properties or operational parameters.

Real-time Inference and Stuck Pipe Detection. Once trained, models provide live predictions of drilling performance metrics and risk indicators. A key use case is stuck pipe detection using a digital twin concept: comparing predicted vs. actual mechanical behavior to identify deviations that signal dysfunction. Although validation is still in progress, early testing shows encouraging results. This pipeline is designed for low-latency operation, enabling real-time alerting and integration with downstream services such as reporting and operator chat interfaces.

Fig. 2. illustrates a real-time dashboard for stuck pipe detection using machine learning models for surface torque, weight on bit (WOB), and standpipe pressure (SPP). The top panel displays surface torque deviations, where periods of high trend and value deviation are visually marked in red, indicating potential stuck pipe conditions. The lower panels show corresponding deviations in SPP and WOB, with sharp spikes in trend and value deviations near the same time intervals, reinforcing the anomaly. The system runs three independent models in parallel—Surface Torque, WOB, and SPP—and dynamically highlights critical drilling states, offering early warnings based on learned patterns from historical and simulated data.

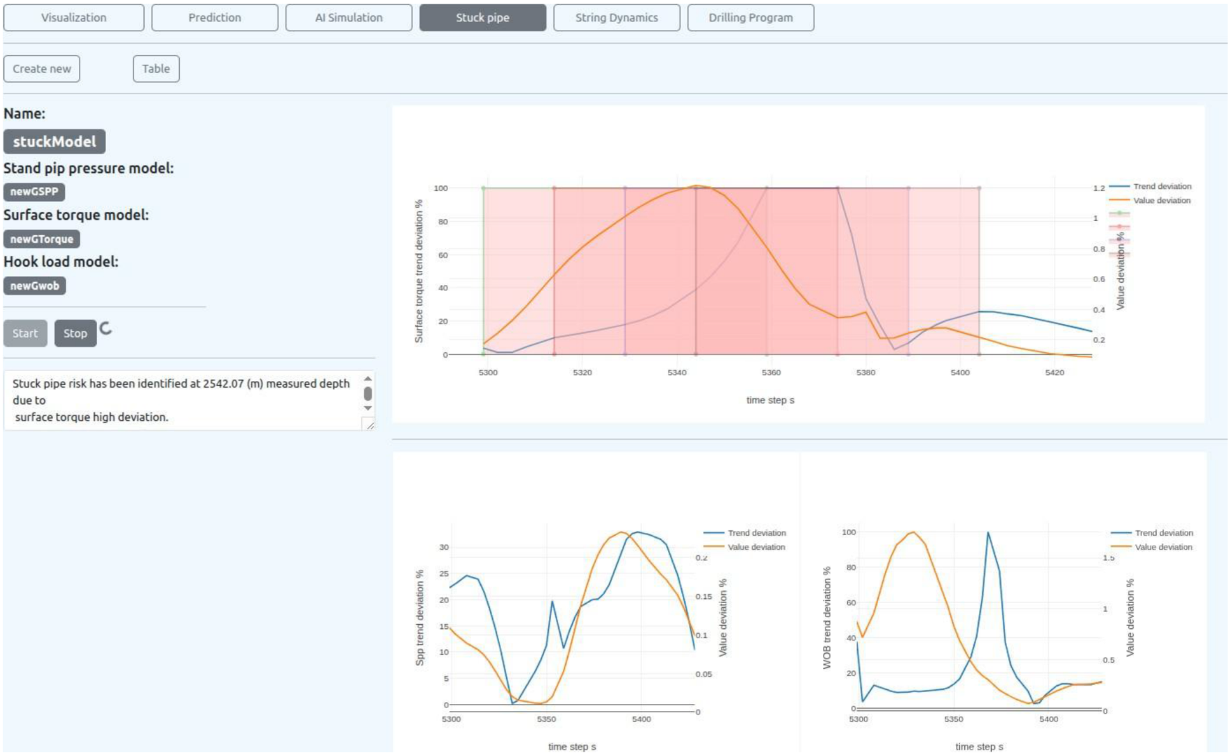

Figure 2—Dashboard visualization of real-time stuck pipe detection using digital twin methodology. Top: surface torque deviation flagged as a high-risk zone (highlighted in red). Bottom left: standpipe pressure (SPP) deviation; bottom right: weight on bit (WOB) deviation. Deviations are calculated by comparing predicted values from machine learning models with actual sensor data.

LLM Integration and Generative AI Agents

To enhance user interaction, automate text processing, and support decision-making, the platform integrates large language models through a dedicated LLM Agent Service. These Generative AI components enable natural language understanding, information extraction, and intelligent prompt-based workflows tailored to drilling operations. The model combines LLMs with vector-based semantic search to generate responses grounded in external documents, thereby enhancing accuracy and context (Lewis et al. 2020).

Fig.3. illustrates the Retrieval-Augmented Generation process architecture, highlighting five key functional steps in enhancing language model outputs with contextual information. (1) The process begins with the user submitting a Prompt + Query to the interface. (2) The system then parses the query and initiates a semantic Search for relevant Information from external knowledge sources, such as document databases or APIs. (3) Retrieved contextually relevant data is returned and used to enrich the original query into an enhanced context package. (4) This enriched prompt—including the original user input and the retrieved knowledge—is then sent to the LLM endpoint for inference. (5) The LLM uses this comprehensive input to generate a more accurate and informed text response, which is delivered back to the user. This approach significantly improves the factual consistency and depth of Generative AI outputs by grounding them in external domain-specific information.

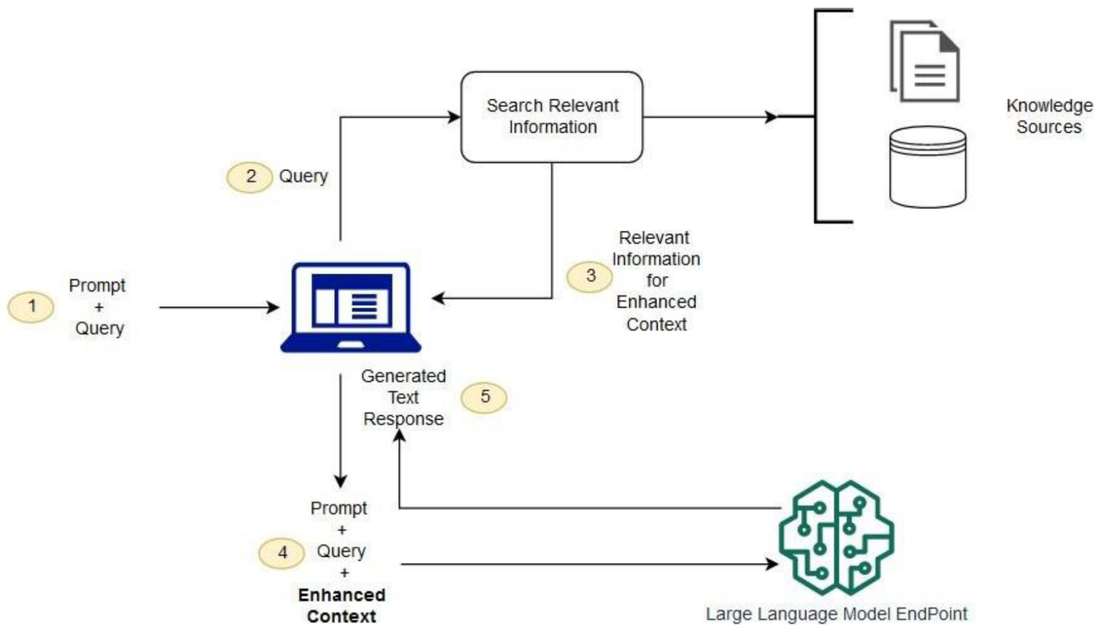

Figure 3—Retrieval-Augmented Generation architecture. The model combines LLMs with vector-based semantic to generate responses grounded in external documents, enhancing accuracy and context (Amazon Web Services 2024).

Agent Design and Prompt Engineering. LLM agents are designed using prompt chaining and tool integration principles. Prompts are engineered for specific tasks, including:

- Extracting drill string specifications and control parameters (e.g., flow rate, RPM, mud weight) from unstructured drilling reports.
- Summarizing operation outcomes and model predictions.
- Responding to natural language queries from drilling engineers via a chat interface.

Agents operate over structured knowledge bases or unstructured PDFs using retrieval-augmented generation, backed by a vector database populated with drilling documents, daily reports, and procedural texts.

LLM-Assisted Simulation Setup. A notable demonstration use case is the LLM-assisted drill string dynamic simulation, applied here in an offline "what-if" mode rather than as a core operational tool. In this setup, agents parse textual reports to identify input parameters such as pipe length and bottomhole assembly (BHA) configuration, then automatically feed them into the simulation module. This capability reduces manual effort, improves repeatability, and makes scenario-based simulations more accessible to non-specialist users, while the platform's primary function remains real-time monitoring and continuous model adaptation.

Report Generation and Chat Services. LLMs are also used for auto-generating technical summaries, daily reports, and real-time status updates. These outputs are presented in human-readable form, improving communication between field operators and engineers. Overall, LLM agents improve usability, reduce manual processing, and introduce intelligent assistance throughout the drilling workflow.

Drill String Dynamics Simulation Module

The platform also provides access to a drill string dynamics simulation module, used here as an optional, offline tool for validation and scenario-based testing rather than as a core operational component. This module models the mechanical behavior of the drill string under various downhole conditions and operates as a standalone microservice, allowing it to be run independently when needed. See Fig. 4.

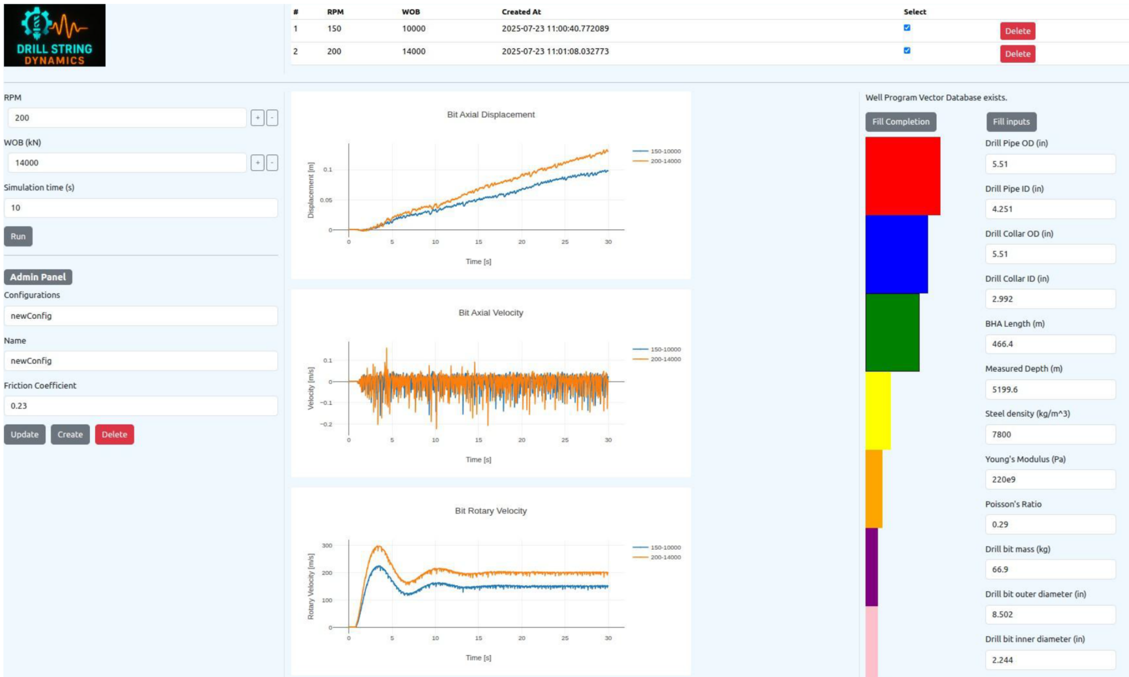

Figure 4—Drill String Dynamics simulation interface with LLM-assisted configuration. The interface allows users to input drill string parameters (e.g., RPM, WOB, pipe geometry, material properties) and visualize results such as bit axial displacement, axial velocity, and rotary velocity. Multiple configurations can be run and compared, supporting analysis of dynamic behavior under different operating conditions.

Fig. 4. presents the graphical user interface (GUI) of the drill string dynamics simulator, developed to evaluate the axial and torsional behavior of the drill bit under varying operational conditions. On the left panel, users can input key drilling parameters such as rotary speed (RPM), weight on bit, and simulation time, which serve as inputs to simulate different what-if scenarios. Upon clicking "Run" the simulator computes and visualizes the bit's axial displacement, axial velocity, and rotary velocity over time, allowing engineers to assess dynamic responses under specific conditions. The system also features an Admin Panel, where users can configure and store multiple simulation setups, including settings like friction coefficient and custom naming conventions for reuse. On the right side of the GUI, the application supports automatic extraction of well parameters from a vectorized drilling program. By clicking "Fill Completion" well sections will be extracted. Then, a section will be selected by the user, and by clicking "Fill Inputs" the simulator searches the drilling program vector database and auto-populates relevant input fields—such as drill pipe dimensions, bottom hole assembly length, depth, and material properties—which are then fed directly into the drill string dynamics model. This integration provides a seamless and intelligent interface for evaluating how changes in surface parameters influence drill string performance, enhancing the ability to prevent dysfunctions such as stick-slip, whirl, or excessive vibrations in real time.

Simulation Inputs and Modeling. The simulator takes as input a range of mechanical and operational parameters, including:

- Drill string geometry (pipe length, diameter, wall thickness)
- Bottom Hole Assembly configuration
- Control parameters such as surface RPM, weight on bit, and fluid properties

The model computes dynamic responses such as axial vibration, torque and drag, and stick-slip behavior. These outputs are used to compare predicted mechanical responses with those observed in the real-time data stream or to simulate operational changes.

LLM-Assisted Configuration. To streamline simulation setup, the platform uses LLM agents to extract required parameters from unstructured drilling reports. These agents identify relevant text spans, extract numerical values, and convert them into structured JSON input for the simulator. This allows field engineers to run complex simulations by simply uploading a report or entering a query, significantly reducing setup time and human error. See Fig. 4.

Usage and Current Scope. The drill string dynamics simulation module is currently used for manual scenario analysis based on basic if-else logic or operator requests. Simulation results can be saved, visualized, and compared via the front end. Combined with real-time data and prediction outputs, the simulation module contributes to a richer, multi-layered view of drilling performance and operational risk.

Application environment

This section details the multi-functional capabilities of the AI-enhanced, microservice-based environment developed for real-time drilling support, with emphasis on real-time monitoring, visualization, and LLM-agent collaboration, while including simulations as an optional tool for scenario-based demonstrations.

Real-Time Simulations

The application integrates multiple simulation environments, including a web-based drilling simulator (Saadallah et al. 2019), enabling seamless data exchange via RESTful APIs (Masse 2011). Real-time parameters such as WOB, ROP, hookload, and standpipe pressure are streamed and visualized with interactive control capabilities, allowing users to inject set points in a closed-loop feedback system (Rommetveit et al. 2007). See Fig.5.

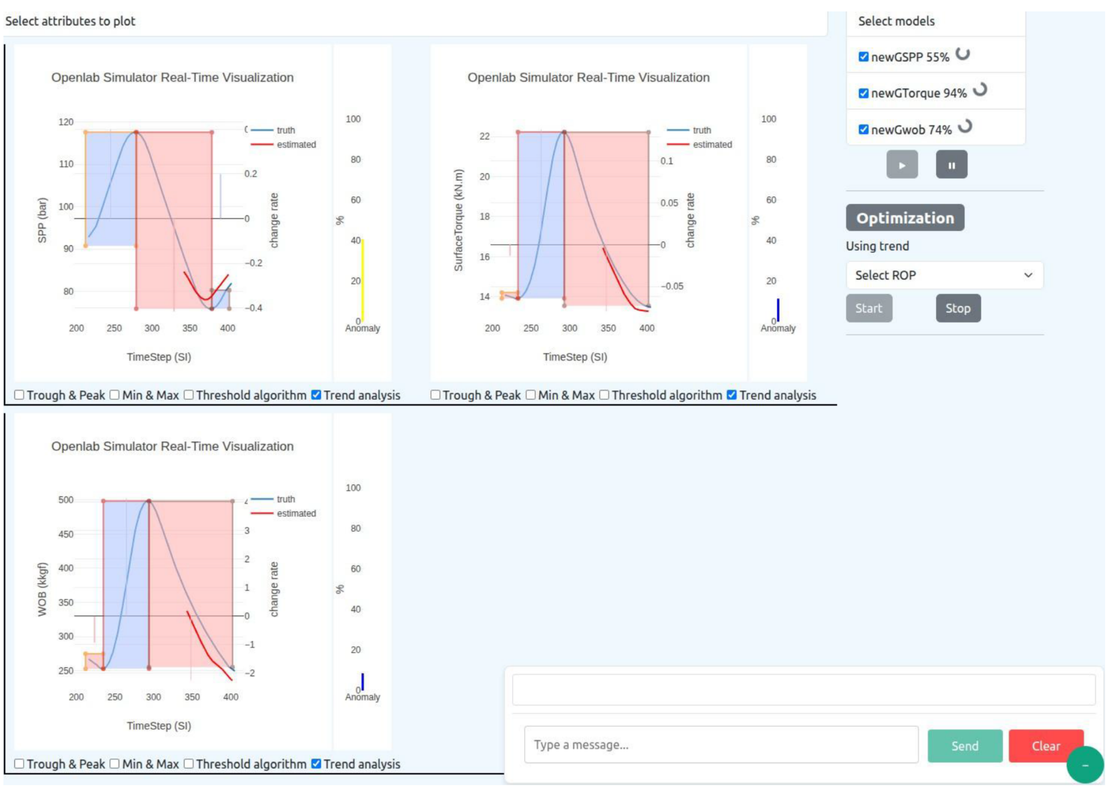

Figure 5—Real-time monitoring dashboard of surface parameters from a web-based drilling simulator (Saadallah et al. 2019) integrated with AI model predictions. The plots show estimated vs. measured values for Standpipe pressure (SPP), Surface torque, and Weight on bit (WOB), with anomaly detection based on deviation thresholds and trend changes. Blue and shaded regions highlight periods of uprising and down rising, respectively.

Fig. 5 displays a real-time monitoring dashboard connected to a web-based drilling simulator (Saadallah et al. 2019), showcasing surface parameters—Standpipe pressure, Surface torque, and Weight on bit. For each parameter, the dashboard visualizes both actual measurements (blue line) and trend predictions from trained ML models (red line). Deviation trends and change rates are overlaid to assess behavioral dynamics, and a vertical anomaly bar indicates anomaly levels from 0 to 100%, where higher values reflect larger deviations between predicted and observed trends.

The panel on the right includes model statuses such as "newGSPP 55%", indicating the current confidence or performance level of each model based on real-time input. The "Optimization" module shown here is a simple trend-based controller designed to improve drilling efficiency. Specifically, it monitors the ROP trend, and if a declining trend is detected, it automatically increases WOB incrementally to counteract the drop and boost ROP performance. This creates a feedback loop using trend direction as a guide for corrective actions.

Moreover, this interface supports interaction with a chatbot assistant, enabling engineers to query system performance, request parameter summaries, or investigate anomaly causes using natural language. The chatbot integrates LLM reasoning with real-time model outputs, enhancing decision-making through intuitive, context-aware QA.

Visualization of Machine Learning Results

Real-time data streams are processed and visualized across the dashboard interface with capabilities that include:

- Multi-model comparison in parallel,
- Custom feature highlighting (e.g., peaks, thresholds),
- Support for evaluating model performance during training,
- Visualization of model evolution with changing inputs and retraining dynamics.

See Fig. 6.

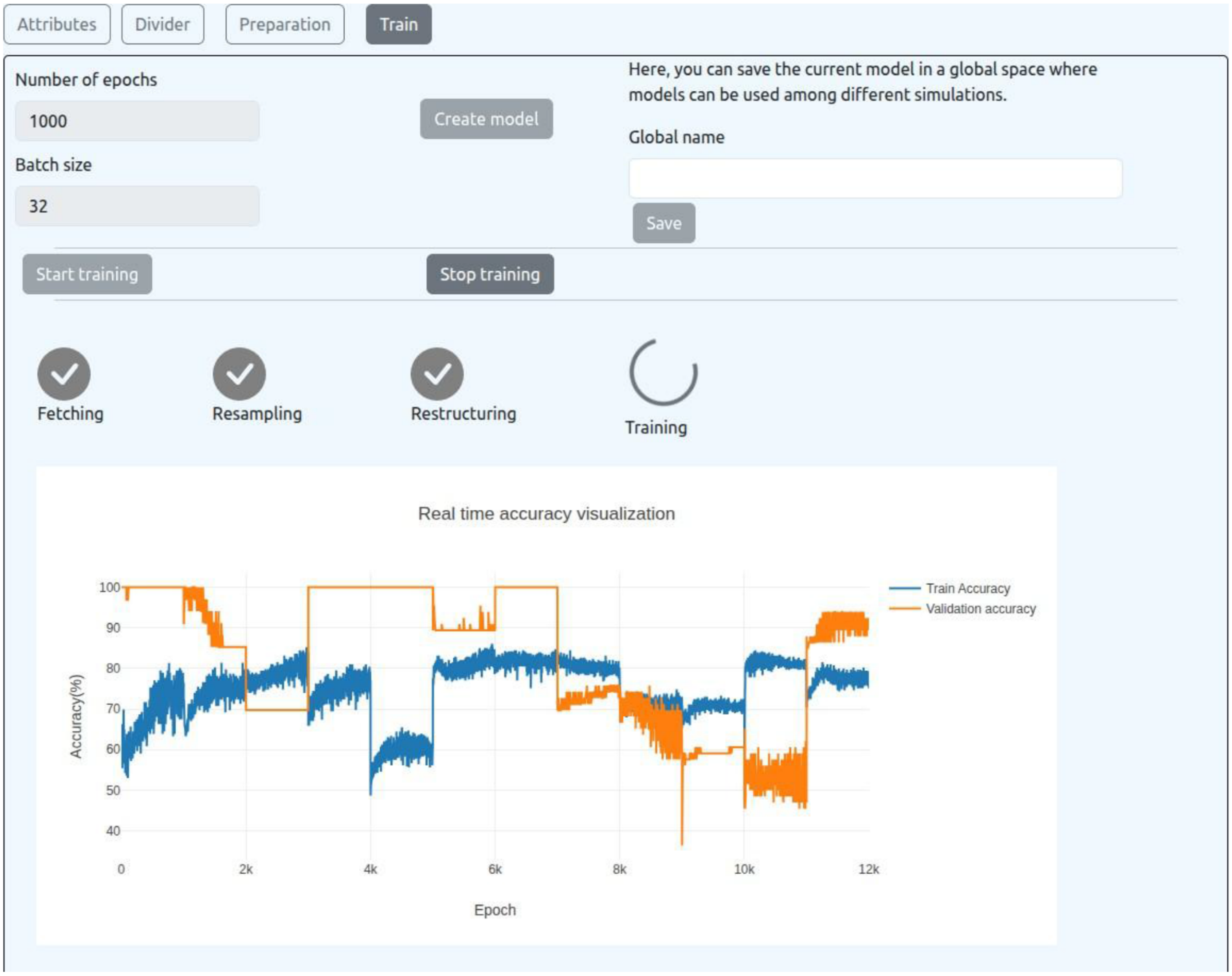

Figure 6—Real-time training dashboard for drilling performance models within the AI platform. The interface enables continuous machine learning during operations, allowing the user to configure parameters such as the number of epochs and batch size, and initiate training. The plots track training and validation accuracy across 12,000 epochs, with system status indicators reflecting progress through data fetching, resampling, restructuring, and ongoing training steps.

Continuous Training Model Interaction

A key novelty is the ability to train machine learning models continuously using incoming data, while simultaneously displaying intermediate model outputs. Users can select features, modify training parameters, and view performance metrics live, helping mitigate issues like overfitting or concept drift in dynamic drilling environments (Sahebi and Sui 2024).

Fig. 6. presents the real-time training module for drilling performance models, enabling continuous machine learning during operations. The user can configure parameters such as the number of epochs and batch size, and initiate training with a simple interface. The training process involves sequential steps—fetching, resampling, and restructuring—before entering the active training phase. Importantly, the model is trained continuously using newly arriving data batches, allowing it to dynamically adapt to changing drilling conditions in real time. This ensures that predictions remain accurate and context- aware throughout the operation. The graph visualizes both training accuracy and validation accuracy, helping the user assess model generalization. Training proceeds until manually stopped by the user or automatically paused when real-time accuracy reaches a predefined threshold, preventing overtraining. The resulting model can then be saved globally for reuse across multiple simulations. This capability supports adaptive learning, crucial for modern data-driven drilling systems.

LLM-Assisted Interfaces and Chat Agent Integration

The environment includes a chat-based interface powered by an LLM agent that allows users to:

- Generate technical reports automatically. See Fig. 7. and Fig. 8.
- Interpret sensor data in natural language.
- Trigger model development pipelines via conversation.
- Request specific visualizations or historical model diagnostics.

This assistant is further integrated into drill string dynamics interpretation, helping engineers by summarizing dynamic events (e.g., stick-slip) detected via signal analysis and AI classifiers.

Conclusion

This research presents a modular, AI-driven platform for real-time drilling performance monitoring, demonstrated using hybrid data from a web-based drilling simulator and replayed historic field datasets. Built on a robust microservice architecture, the system combines continuous machine learning, large language models, and semantic document search to support adaptive, intelligent operations.

Key innovations include real-time data ingestion, automatic retraining of predictive models, and early incident detection—such as stuck pipe risks—through deviation-based anomaly analysis. A built-in LLM agent enables natural language interactions, automates report generation, and allows engineers to query drilling documents conversationally using a vector database.

Scenario tests using replayed field and simulated data demonstrated the system's ability to maintain model accuracy under dynamic conditions, reduce manual workload, and deliver early warnings for operational anomalies. Its flexible architecture ensures scalability and smooth integration into existing infrastructure—on-premises or cloud. This work represents a pioneering step toward autonomous drilling systems by unifying AI-powered insights, explainable decision support, and human-friendly interfaces in a single platform.

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Appendix

Figs 7. and 8. present an overview of surface parameter deviations—Standpipe pressure, Weight on bit, and Surface torque—plotted against measured depth, with each parameter shown as both absolute deviation (Fig. 8) and trend-based deviation (Fig. 7). These visualizations provide insight into potential stuck pipe indicators during drilling.

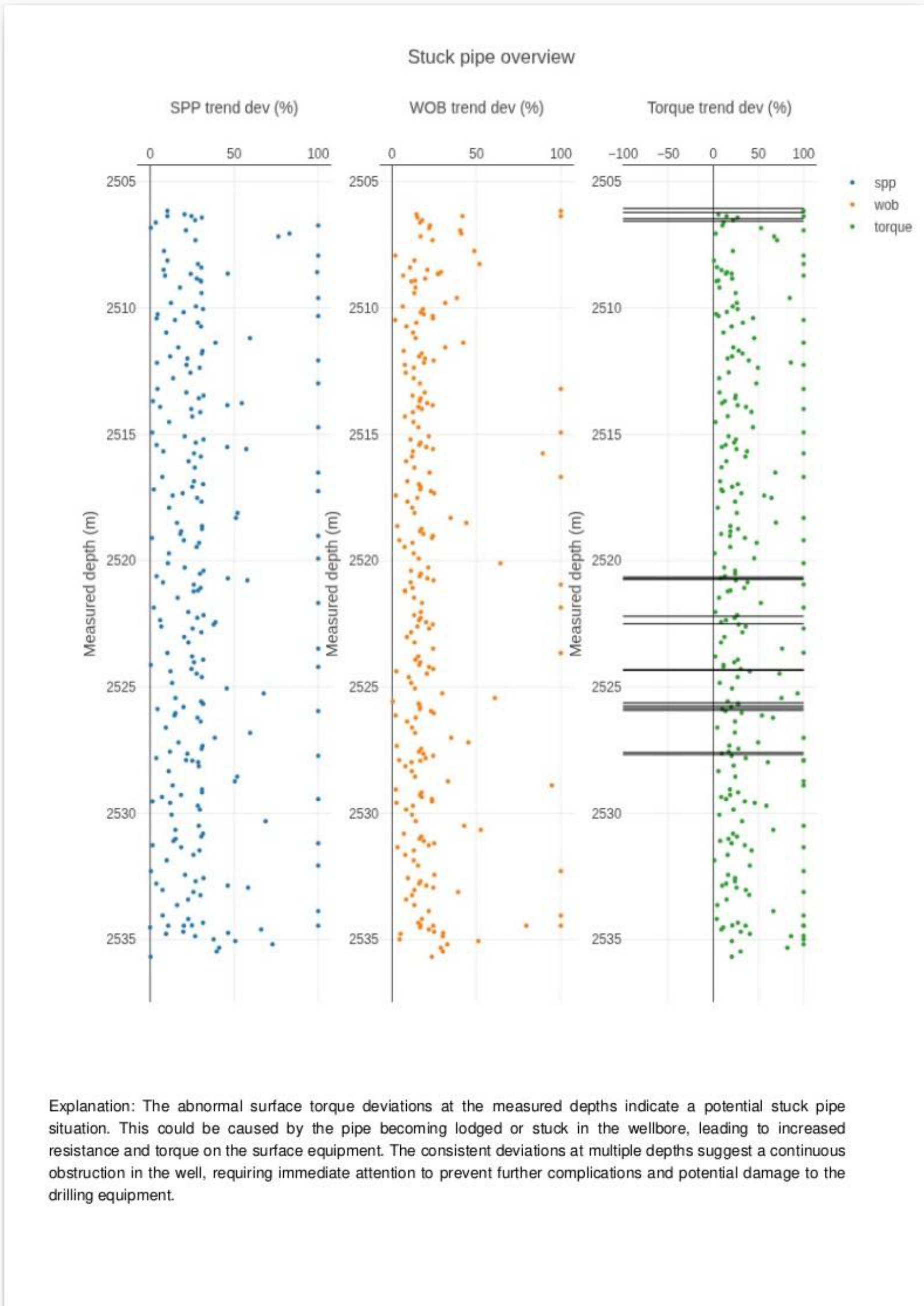

Figure 7—Stuck pipe detection using trend deviation analysis. Surface torque anomalies are detected at multiple depths, with automated explanation generated by LLM agents based on real-time model outputs.

In Fig. 8, absolute deviations are calculated as percentage changes from expected values, highlighting localized anomalies. Notably, deviations in WOB and SPP around depth 2525 m suggest abnormal mechanical responses, which may indicate increased drag or resistance. Figs 7. expands on this by showing trend-based deviations, which capture the evolving behavior of each parameter. The persistent and significant deviations in surface torque across a range of depths strongly point to a potential stuck pipe event. These combined views help identify both instantaneous spikes and consistent patterns over depth, offering a comprehensive diagnostic tool to detect and localize downhole dysfunctions early in the drilling process.

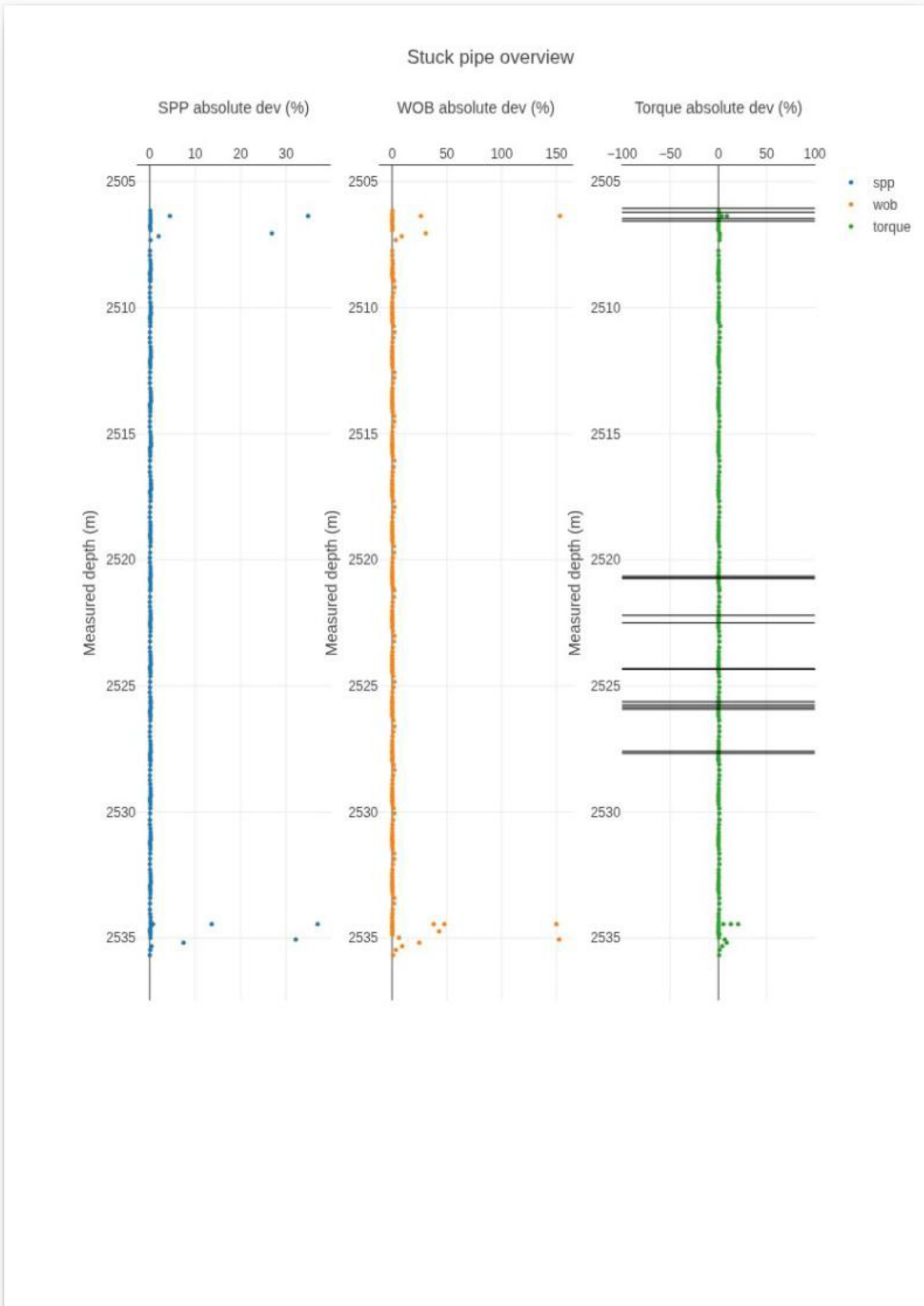

Figure 8—Stuck pipe detection using absolute deviation analysis of surface torque, WOB, and SPP across measured depths. Highlighted anomalies indicate critical zones with significant deviation magnitudes, suggesting potential pipe sticking risk.